<Kh.Kim / >

Summary

Software Engineer with 7 years of experience.

My work started in HCI and Visual Analytics, where I studied how people understand and work with complex systems. That interest has continued through AutoML and explainable AI (XAI), data visualization, content platforms, server-driven UI, and internal operations automation.

Recently, I have been interested in using AI to help people make better decisions with less effort. Through GitHub-centered collaboration, browser automation, and executable prototypes, I am exploring ways to make scattered information and evidence easier to work with.

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Themes

Human-System Interaction

I came from HCI and Visual Analytics research. The work is making a model's output something a person can actually interrogate on screen.

Understandable AI Systems

I built the prediction-explanation UI for an AutoML product. Without the reasoning beside the number, a non-expert has no basis for acting on it.

Product Experimentation Infrastructure

Server-driven UI let us change screens and run A/B tests without waiting on a release. Existing screens moved over gradually rather than in one rewrite.

AI-Native Development Workflow

I would rather show a working screen than describe an idea in a document. Pairing with an agent shortens that gap to a day or two.

Collaboration and Operations Tooling

Every tool keeps its own context, which leaves a person to carry state between them — what the issue said into the PR, what the log showed into the thread. That carrying belongs in tooling: GitHub workflows, internal-tool integration, browser automation.

Experience

Levvels

Frontend Engineer
Jul 2025 – Present

Exploring AI-native product development and internal operations automation while building creator-commerce frontend systems. I use GitHub-centered collaboration, executable prototypes, and browser automation to reduce the cost of product changes and repeated investigation work.

React Next.js TypeScript TanStack Query Zustand SSE Playwright Sentry Mixpanel ChannelTalk PageAgent Generative UI

AI Character Chat Service

May 2026 – Present

I am building the user-facing AI character chat experience and an Admin for authoring and reviewing character content. The product is being prepared for release.

- Separated in-progress streamed responses from persisted chat history and handled send, completion, failure, and regeneration states
- Built authoring and review tools for stories, lorebooks, search keywords, and other character content
- Resolved differences across product specs, Figma, and Swagger and documented the implementation decisions and QA cases

Vuddy Commerce

Jul 2025 – Present

Built and maintained Vuddy’s customer web, Creator Studio, and internal Admin. Worked across the commerce flow from class discovery and checkout to orders, cancellations, returns, and exchanges.

- Built class discovery, video, and checkout flows along with Studio order management and Admin tools for products, students, and passes
- Updated the cart, identity verification, secondary-creator onboarding, and cancellation, return, and exchange flows as policies and APIs changed
- Fixed HLS home-carousel and mobile WebView issues involving safe areas, scrolling, and back navigation

PageAgent-based Internal Operations Automation

Mar 2026 – Present

Case study

Turned repeated CS/QA investigation work into an automation target, connecting natural-language requests to browser actions and verifiable result UI.

- Connected account, payment, and mapping data from several internal tools so operators could compare the necessary context in one result view
- Extended Alibaba PageAgent for a Chrome Managed Profile environment to automate tab navigation, clicks, and inputs
- Separated LLM output from reproducible action sequences and result presentation
- Simplified repeated CS/QA investigation work that required manually checking multiple tables and services

RIDI Corp.

Frontend Engineer
May 2022 – Jul 2025

Focused on reducing the dependency between product experimentation, operational UI changes, and deployment cycles in a large content platform. Through server-driven UI and gradual migration, I helped create structures that let operators adjust and test experiences faster.

Next.js React TypeScript Emotion Jest PHP Twig Sentry Virtualization

RIDI Web Platform 🔗

May 2022 – Jul 2025

Case study 🔗

Used server-declared UI and island architecture to gradually move legacy pages into React while lowering the execution cost of operational UI changes and A/B tests.

- Rendered server-declared UI on the client to decouple screen changes from deployment cycles
- Migrated PHP pages toward React incrementally to reduce migration risk
- Created a more flexible foundation for A/B tests and operational UI changes
- Maintained stability across mobile, desktop, and special-device environments

Large-scale List Rendering Optimization

May 2022 – Jul 2025

Case study 🔗

Improved list rendering structure and loading experience so large-scale content exploration would not limit user experience or product experimentation.

- Analyzed frame drops from large DOM trees
- Rendered only the visible range with windowing
- Handled scroll position and dynamic heights
- Improved skeleton UI for perceived speed

User Environment Stabilization

May 2022 – Jul 2025

Reduced instability caused by browser, device, and extension differences, improving both product experience and operational response.

- Tracked real-user errors with Sentry
- Handled translation-plugin conflicts
- Improved E-ink rendering and UX issues

TmaxEnterprise

Research Engineer
Feb 2020 – May 2022

Currently TmaxBizAI
Formerly TmaxData

Designed product flows, visualizations, and interactions that helped non-expert users work with complex AI and analytics systems. The focus was not technology first, but the usability, explainability, and workflow of AI systems.

React TypeScript Material-UI Sass Python jQuery

Legacy System React Migration

Jan 2021 – May 2022

Moved a tightly coupled internal UI library toward React to create a more consistent and extensible foundation for platform interfaces.

- Modularized legacy UI code and improved interface structure
- Developed reusable components to replace bespoke legacy widgets

AutoML Platform

Feb 2020 – May 2022

Case study

Designed and implemented codeless studio flows and explainable AI visualizations so users without model-development experience could explore and operate AI capabilities.

- Implemented a codeless studio for non-expert model builders
- Developed explainable AI visualizations and dashboards
- Researched interfaces between AutoML engines and model developers
- Clarified requirements for an early-stage platform

Open Source

OpenMMO

Contributor
2026.07 – Present

Contributing to an open-source MMORPG where AI agents and humans play through the same protocol (Rust · Svelte · Three.js · WebAssembly). Game development was new to me; I started with client tests and now work mostly on the Rust server, and all ten PRs have been merged.

TypeScript Svelte Rust WebAssembly Vitest

Client

- Built the HUD minimap after a written design proposal turned a hesitant maintainer around, reusing the existing terrain bakes and rotation convention so the server needed no change. Following review from the party system's author, dropped the markers that would have kept the client polling and fixed the missing store update found along the way at its source (#11, #77)
- Filled test gaps around fishing state transitions, bilinear water-surface sampling, and click-intent routing. An out-of-range water click now walks toward the target, with the server constant exposed through WebAssembly so both sides read one value, and scheduled sounds are cancelled when the session ends (#56, #64, #65)

Server

- Bounded terrain tile caches that only ever grew, sweeping by generation under a capacity fuse while leaving the read path lock-free (#68)
- Built account-level ban and unban for operators, cutting sessions by account rather than character so an alt cannot walk back in, and closing the window where a connection past the check still registers (#95)
- Refused trade with a sleeping merchant: her sleep schedule is agent-side data the server never sees, so the bed-occupancy interaction it does see stands in for sleep. The check sits in the one validator every trade path already passes through, which also stops a shop window opened before bedtime from charging the buyer (#103)
- Added a night merchant who opens a stall after dark, since the town's only shop sleeps and nothing was open at night. A merchant is defined across two files, so a test now catches the half-added case that would leave a shop no agent can run (#111)

Docs & onboarding

- Documented the terrain bake step in the dev setup after hitting the black-world onboarding gap myself (#78)
- Wrote a CONTRIBUTING guide gathering what new contributors learned the hard way — the checks CI runs, which lived only in the workflow file, and the pre-start checks that avoid duplicate work — linked from all three README translations (#118)

vite-plus

Contributor
2026.08

Contributing to vite-plus, the toolchain from VoidZero (the team behind Vite and Vitest), a Rust · TypeScript hybrid. Started with a merged shell-integration fix on the Rust side; now working on the vite-task task runner's cache on the TypeScript side, alongside issue analysis and PR review across both repos.

Rust TypeScript Shell

Shell integrations

- Generated shell wrappers didn't recognize a vp env use that followed the global -C <dir> flag (-C <dir> · -C<dir> · -C=<dir>), so the environment change never applied to the current shell session; fixed in setup.rs. Also made vpr completions preserve the working-directory option across zsh, Fish, Nushell, and PowerShell, and pinned the four shells' integration templates with snapshot tests. Review rounds added set -u/NO_UNSET hardening of the generated scripts and a PowerShell type-safety fix, each covered by executable regression tests (#2508, issue #2056).

Task runner cache (vite-task)

- Tracked bulk env queries were validated against the full unfiltered environment, so per-run CI variables such as ACTIONS_ORCHESTRATION_ID invalidated the cache on every GitHub Actions run. Extended the existing untrackedEnv contract to bulk queries — filtered when recording and validating, served unchanged — with unit and e2e snapshot coverage (vite-task#693, in review, maintainer-filed issue vite-task#505).

Analysis & review

- Root-caused the task runner not exposing npm lifecycle env to package.json scripts and showed why a vite-plus-side fix can't reach cache-enabled tasks (vite-task#692). Reviews on vite-plus surfaced the same env-fingerprint filtering silently stripping the proposed lifecycle-env fix (#2385) and three verified edge cases in the Oxlint/Oxfmt config migration (#2483).

Education

Ulsan National Institute of Science and Technology (UNIST)

Master of Science in Computer Science
2018.03 – 2020.02
Bachelor of Science in Technology Management (Multi-disciplinary major in Computer Science)
2013.03 – 2018.02

AI-driven UI and Recommender System Research

2019.01 – 2020.02

Case study

Studied how recommender systems should communicate algorithmic exploration so users can understand the system and provide higher-quality feedback.

- Measured the impact of transparency on data quality in the explore-exploit problem of recommendation systems
- Analyzed how disclosure patterns for exploratory items affect user perception and data quality
- Proposed a metric to evaluate the value of user logs from an AI perspective
- Implemented a web-based movie recommendation system for experimental environments
- Designed data-driven UI experiments to observe interaction between people and recommendation algorithms

Python | Flask | Surprise | jQuery

Modeling Web User Behavior

2019.02 – 2020.02

Modeled decision-making and behavior patterns in complex web environments to better understand interaction between people and systems.

- Estimated user reward functions from behavior logs through Inverse Reinforcement Learning
- Dealt with various web environments such as data analysis tools like Tableau, history education tools based on Wikipedia documents, and card games
- Utilized data analysis tools such as Tableau and implemented a history education tool based on Wikipedia documents

Python TensorFlow Keras scikit-learn

Publications

An Empirical Analysis on Transparent Algorithmic Exploration in Recommender Systems 🔗

Kihwan Kim
A Computing Research Repository (CoRR), 2108.00151, 2021

- Investigated how to deliver random items to capture user preferences in recommendation systems
- Implemented a web-based movie recommendation system used as an experimental environment
- Proposed a metric to evaluate the value of user logs from an AI perspective
- Measured the impact of transparency on data quality in the explore-exploit problem of recommendation systems
- Recruited 94 participants from Amazon MTurk
- Collected real usage log data and survey responses by having participants use a Netflix-like experimental environment

ST-GRAT: A Novel Spatio-temporal Graph Attention Networks for Accurately Forecasting Dynamically Changing Road Speed 🔗

Cheonbok Park, Chunggi Lee, Hyojin Bahng, Yunwon Tae, **Kihwan Kim**, Seungmin Jin, Sungahn Ko, Jaegul Choo
ACM International Conference on Information and Knowledge Management (CIKM), 2020

- Preprocessed vehicle detection sensor data installed on the road surface by Korea Expressway Corporation
- Categorized cases where attention worked effectively by patterns

Modeling Exploration/Exploitation Decisions through Mobile Sensing for Understanding Mechanisms of Addiction 🔗

Kihwan Kim, Sanghoon Kim, Chunggi Lee, Sungahn Ko
ACM International Conference on Mobile Systems, Applications, and Services (MobiSys), 2019

- Proposed a system to detect addiction disorders from smartphone usage logs through Inverse Reinforcement Learning

An Empirical Study on the Relationship Between the Number of Coordinated Views and Visual Analysis 🔗

Juyoung Oh, Chunggi Lee, Hwiyeon Kim, **Kihwan Kim**, Osang Kwon, Eric D. Ragan, Bum Chul Kwon, Sungahn Ko
A Computing Research Repository (CoRR), 2204.09524, 2018

- Conducted experiments to see how the number of visualization charts affects data visual analysis
- Had 44 participants use a visual analysis tool and solve data analysis tasks
- Categorized users' analysis patterns through think-aloud protocol, recorded screens, and log data
- Observed a positive correlation between the number of charts and task scores

A Survey of Visualization Techniques for Interpreting Deep Learning 🔗

Jaesung Lee, **Kihwan Kim**, Chunggi Lee, Sungahn Ko
Noise & Vibration, Vol.27 No.6, 2017.11

- Summarized and investigated visualization techniques for interpreting deep learning models